

Project **ExploreConnectSA - Travel Booking System**

Group SD8

Module **ADP3 (Application Development Practice 3)**

Milestone **Factory Implementation**

Submission Date **July 2026**

	Student Number	Role
Zamandlovu C Ndlovu	211204803	Group Leader / Lead Developer
Alakhe Sbusiso Mxakato	230485316	Developer
Entle Myezo	230076238	Developer
Kabelo Nimrod Moloko	230117015	Developer
Ricardo Mukwevho	222567023	Developer
Somila Ndobozza	231157592	Developer

**The report of the Factory Implementation milestone for the ExploreConnectSA
Travel Booking System.**

Due to unforeseen circumstances including the school holiday period and internet connectivity challenges faced by group members who returned to their homes, the project experienced delays. As the Group Leader, I (Zamandlovu C Ndlovu) stepped in to ensure the project remained on track and was completed successfully.

Challenges Face

Challenge	Impact	Resolution
School Holidays	Team members unavailable during critical development period	Group Leader stepped in to complete pending tasks
Internet Connectivity Issues	Remote team members unable to contribute effectively	Group Leader took over development tasks
Time Constraints	Project at risk of missing deadline	Focused effort on completing all factory classes
Coordination Challenges	Difficulty in task assignment and follow-up	Centralized task management by Group Leader

Technical Challenges

Challenge	Solution
Null values in CreditCardDetails Factory	Implemented automatic calculation of lastFourDigits and cardType in Builder
InvoiceFactoryTest failing due to zero subtotal	Fixed by setting subtotal and taxes in the test setup
PaymentDetailsFactoryTest expecting PAID status	Added processPayment() call before assertions
ReviewFactoryTest requiring non-null Booking	Created a valid Booking in the test setup
Email sending test failure	Added null checks in sendViaEmail() method

Completed Factory Classes

Alakhe Sbusiso Mxakato (230485316) - Completed Tasks

	Class	Status
1	TravelerFactory.java + TravelerFactoryTest.java	Complete
2	ReviewFactory.java + ReviewFactoryTest.java	Complete
3	LoyaltyProgramFactory.java + LoyaltyProgramFactoryTest.java	Complete
4	ReportFactory.java + ReportFactoryTest.java	Complete by Group Leader

Group Leader - Completed Tasks (Originally Assigned to Kabelo Moloko)

	Class	Original Assignee	Completed By
5	ShuttleBookingFactory.java + Test	Kabelo Moloko	Group Leader
6	HotelBookingFactory.java + Test	Kabelo Moloko	Group Leader
7	FlightBookingFactory.java + Test	Kabelo Moloko	Group Leader
8	CarRentalBookingFactory.java + Test	Kabelo Moloko	Group Leader

Group Leader - Completed Tasks (Originally Assigned to Ricardo Mukwevho)

	Class	Original Assignee	Completed By
9	PaymentDetailsFactory.java + Test	Ricardo Mukwevho	Group Leader
10	InvoiceFactory.java + Test	Ricardo Mukwevho	Group Leader
11	CreditCardDetailsFactory.java + Test	Ricardo Mukwevho	Group Leader

Group Leader - Completed Tasks (Originally Assigned to Ricardo Mukwevho)

	Class	Original Assignee	Completed By
12	LocationFactory.java + Test	Somila Ndobozza	Group Leader
13	LineItemFactory.java + Test	Somila Ndobozza	Group Leader
14	BoardingPassFactory.java + Test	Somila Ndobozza	Group Leader

Group Leader - Completed Tasks (Self-Assigned)

	Class	Author	Status
15	CustomerFactory.java + Test	Zamandlovu C Ndlovu	Complete
16	ContactDetailsFactory.java + Test	Zamandlovu C Ndlovu	Complete
17	AdminFactory.java + Test	Zamandlovu C Ndlovu	Complete
18	AddressFactory.java + Test	Zamandlovu C Ndlovu	Complete

Group Leader - Completed Tasks (Originally Assigned to Entle Myezo)

	Class	Original Assignee	Completed By
19	CancellationPolicyFactory.java + Test	Entle Myezo	Group Leader
20	BookingConfirmationFactory.java + Test	Entle Myezo	Group Leader
21	BillingAddressFactory.java + Test	Entle Myezo	Group Leader

Work Distribution

Team Member	Assigned Tasks	Completed	Completed By	Percentage
Alakhe Sbusiso Mxakato	4	3	Self + Group Leader	90%
Zamandlovu C Ndlovu (Group Leader)	4	18	Self	100%
Entle Myezo	3	0	Group Leader	0%
Kabelo Nimrod Moloko	4	0	Group Leader	0%
Ricardo Mukwevho	3	0	Group Leader	0%
Somila Ndobozza	3	0	Group Leader	0%

Test Execution Results

Test Class	Passed	Failed	Status
AddressFactoryTest	All	0	PASS

AdminFactoryTest	All	0	PASS
BillingAddressFactoryTest	All	0	PASS
BoardingPassFactoryTest	All	0	PASS
BookingConfirmationFactoryTest	All	0	PASS
CancellationPolicyFactoryTest	All	0	PASS
CarRentalBookingFactoryTest	All	0	PASS
ContactDetailsFactoryTest	All	0	PASS
CreditCardDetailsFactoryTest	All	0	PASS
CustomerFactoryTest	All	0	PASS
FlightBookingFactoryTest	All	0	PASS
HotelBookingFactoryTest	All	0	PASS
InvoiceFactoryTest	All	0	PASS
LineItemFactoryTest	All	0	PASS
LocationFactoryTest	All	0	PASS
LoyaltyProgramFactoryTest	All	0	PASS
PaymentDetailsFactoryTest	All	0	PASS
ReportFactoryTest	All	0	PASS
ReviewFactoryTest	All	0	PASS
ShuttleBookingFactoryTest	All	0	PASS
TravelerFactoryTest	All	0	PASS

In Conclusion

The Factory Implementation milestone has been successfully completed with all 21 factory classes and their corresponding test classes fully implemented and passing all tests.

Despite significant challenges including school holidays, internet connectivity issues, and team member unavailability, the project was completed successfully through the

dedication and commitment of the Group Leader who stepped in to ensure all tasks were completed.

The project is now ready to proceed to the next milestone: **Repository Implementation.**